Eric Liu

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Education

University of Toronto **Sep 2022 – May 2027**
Bachelor of Applied Science in Computer Engineering with PEY Co-op Toronto, Ontario

- On track to graduate with Minors in Artificial Intelligence Engineering and Engineering Business
- Dean's List Scholar
- Relevant Courses: DSA, Software Communication and Design, Applied Fundamentals of Deep Learning, Networks

Technical Skills

Programming Languages: Python, C++, C, SQL, Java, JS, Verilog, MATLAB, Assembly, Angular, HTML/CSS
Frameworks & Tools: Stable Baselines3, PostgreSQL, PyTorch, Gym, Git, JSON, Docker, Linux, NumPy, AWS, GCP
Strong skills in advanced DSA: Graph Theory, Dynamic Programming, Monotonic Structures, Greedy Algorithms

Experience

Scotiabank – Quantitative Trading Developer **May 2025 – April 2026**
Electronic Execution Services (Algorithmic Trading) – Global Banking and Markets *Toronto, Ontario*

- **Tech Environment:** SQL, Python, Angular, Pandas, Bloomberg, TCA, Market Making, Reinforcement Learning.
- Completed the Outage App, a Bloomberg-integrated platform that analyzes failed algorithmic executions (e.g., market halts, volume limits, client caps) and computes refund adjustments using trade-specific rates; optimized performance via vectorization and bitmasking to process extended outage windows.
- Improved Transaction Cost Analysis, Foreign Exchange and Commission reconciliation pipelines.
- Developed and automated client-facing analytics tools, including commission trackers and client billings - reducing manual reporting effort across trading desk.
- Collaborated with the Program Trading team on index modeling and rebalancing workflows
- Working on a reinforcement learning driven internal market-making system that intercepts retail order flow before external routing. Improving overall profitability and reducing market impact.

Big Tuna - Software Developer **Aug 2020 – May 2024**
Seasonal Game Developer *Oakville, Ontario*

- **Tech Environment:** JavaScript, JSON, SQL, Developer Portal, Game Design, Git, Software Testing.
- Part of a three-person team that created and maintained a highly engaging Discord bot with more than 23,000 players.
- Integrated SQL database updates corresponding to new content, and conducted comprehensive pre-deployment testing.

Research

Market Making Via Reinforcement Learning Research **May 2023 – Aug 2024**
Research Assistant Under Professor Wei Xu *Toronto, Ontario*

- **Tech Environment:** Python, Stable Baselines (DDPG and DQN algorithms), Linux, Pandas, Gym, PyTorch.
- Researched and developed a profitable market-making agent for cryptocurrency exchange using Deep Deterministic Policy Gradient (DDPG) and Deep Q-Network (DQN), understanding its underlying mathematical and financial constructs.
- Crafted a sophisticated machine learning environment that mimics real-world market conditions and scenarios, integrating over 100 million tradebook data entries, paired with a dynamic reward framework that reacts and gives appropriate response to the model's decisions, facilitating continuous learning and strategic optimization.

Extracurricular

University of Toronto Aerospace Design Team **Jul 2023 – Present**
Liquid Propulsion and Aerodynamics Division in Rocketry *Toronto, Ontario*

- Completing code rework that simulates flight data and predicts results, transitioning an outdated 2D point system with two degrees of freedom (2DoF) to a more accurate 3D model with six degrees of freedom (6DoF).

Ontario Engineering Competition (OEC) **January 2024**
Represented the University of Toronto for programming competition *Kingston, Ontario*

University of Toronto Engineering Kompetition (UTEK) **November 2023**
First place in programming competition (Team of 4) *Toronto, Ontario*

Projects

Fungi Identification Project using Deep Learning | *Python, Git, PyTorch, NumPy, Pandas* **Jul 2024**

- Developed a comprehensive Convolutional Neural Network (CNN) model that identifies various species of fungi, trained on over 10,000 images, achieving model accuracy of 81% compared to baseline accuracy of 65%.

GIS Mapping System | *C++, GTK, Git, Multithreading, Heuristic Algorithms* **May 2024**

- Implemented map functionalities, graphic rendering, and multithreading optimizations, and utilized time precomputation methods to enhance performance, reducing time complexity.